MODULE 1 - Ecosystems (Structure and Function)
Ecosystems (Structure and Function): Forest, Desert, Wetlands, River, Oceanic and Lake. Sustainability: 17 SDGs - History, targets, implementation , Capacity Development

SESSION	Topics	BL	CO
01	Introduction	L2	CO1
02	Ecosystems (Structure and Function): Forest, Desert	L2	CO1
03	Wetlands, River, Oceanic and Lake	L2	CO1
04	Sustainability: 17 SDGs - History, targets	L2	CO1
05	SDGs implementation	L2	CO1
06	Capacity Development	L2	CO1

Introduction

The word Environment is derived from the French word "Environ" which means "surrounding". Our surrounding includes biotic factors like human beings, Plants, animals, microbes, etc and abiotic factors such as light, air, water, soil, etc

Components of Environment

Environment has been classified into four major components:

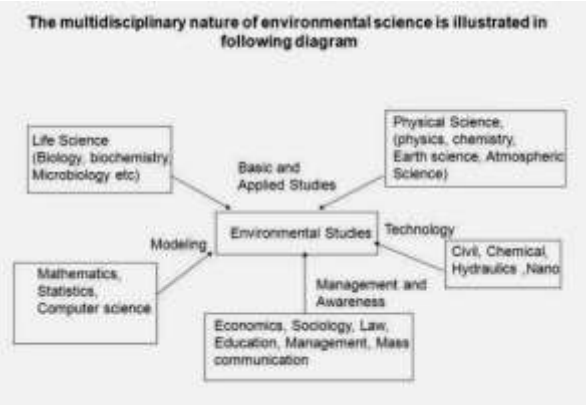
1. Hydrosphere,
2. Lithosphere,
3. Atmosphere,
4. Biosphere.

Hydrosphere includes all water bodies such as lakes, ponds, rivers, streams and ocean etc. Hydrosphere functions in a cyclic nature, which is termed as hydrological cycle or water cycle.

Lithosphere means the mantle of rocks constituting the earth's crust. The earth is a cold spherical solid planet of the solar system, which spins in its axis and revolves around the sun at a certain constant distance. Lithosphere mainly, contains soil, earth rocks, mountain etc. Lithosphere is divided into three layers-crusts, mantle and core (outer and inner).

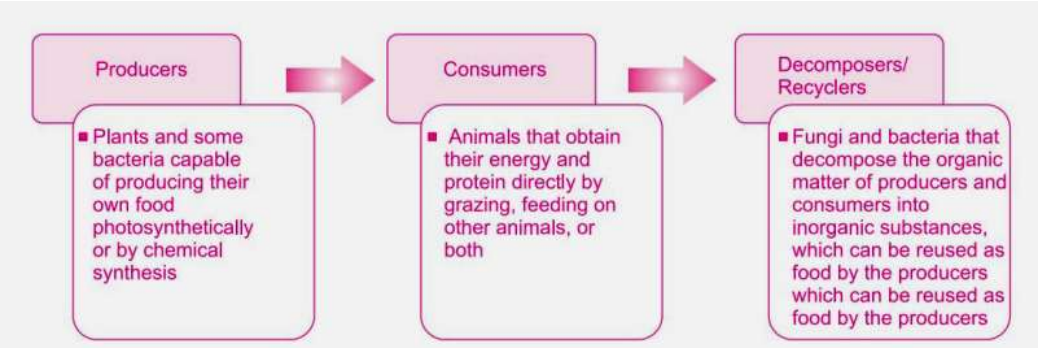
Atmosphere The cover of the air, that envelope the earth is known as the atmosphere. Atmosphere is a thin layer which contains gases like oxygen, carbon dioxide etc. and which protects the solid earth and human beings from the harmful radiations of the sun. There are five concentric layers within the atmosphere, which can be differentiated on the basis of temperature and each layer has its own characteristics. These include the troposphere, the stratosphere, the mesosphere, the thermosphere and the exosphere.

Biosphere it is otherwise known as the life layer, it refers to all organisms on the earth's surface and their interaction with water and air. It consists of plants, animals and micro-organisms, ranging from the tiniest microscopic organism to the largest whales in the sea. Biology is concerned with how millions of species of animals, plants and other organisms grow, feed, move, reproduce and evolve over long periods of time in different environments



Concept of an Ecosystem - Producers, Consumers and Decomposers

Ecology is the study of the relationship between organisms and their environment. An ecosystem is a biotic assemblage of plants, animals, and microbes, taken together with their physico-chemical environment. In an ecosystem the biological cycling of materials is maintained by three groups, viz., producers, consumers, and decomposers/recyclers



Structure and function of an Ecosystem

The structure of an ecosystem is the description of the organisms and physical features of environment including the amount and distribution of nutrients in the ecosystem. From the structure point of view, all ecosystems consist of the following basic components:

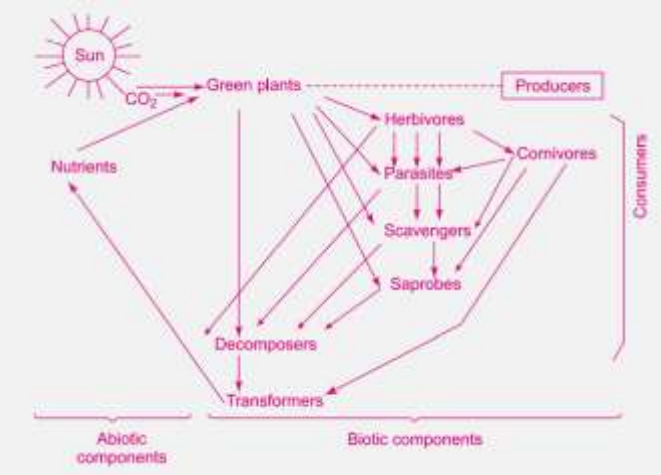
Abiotic Components

They include basic inorganic elements and compounds, such as soil, water, oxygen, calcium carbonates, phosphates and a variety of organic compounds.

Biotic Components

They include all living organisms present in the environmental system which can be further classified into the following three types.

1. Producers (Autotrophic components)
2. Consumers
3. Decomposers



Desert Ecosystems

Each desert is different in many ways; however all deserts receive very low rainfall and are very hot during the day and very cold during the night.

Some deserts are made of very fine, red sand, while others consist of sand mixed with pebbles and rocks. The different components of a desert ecosystem are listed below.

Abiotic Component: Abiotic component includes nutrients present in soil and the aerial environment. The characteristic feature of abiotic component is lack of organic matter in the soil and scarcity of water.

Biotic Component: The various biotic components representing three functional groups are:

Producer organisms: The producers are mainly shrubs or bushes, some grasses and a few trees. The most famous desert plant is the cactus. It can hold water inside its soft tissue.

Consumers The commonly found consumers of a desert ecosystem are following.

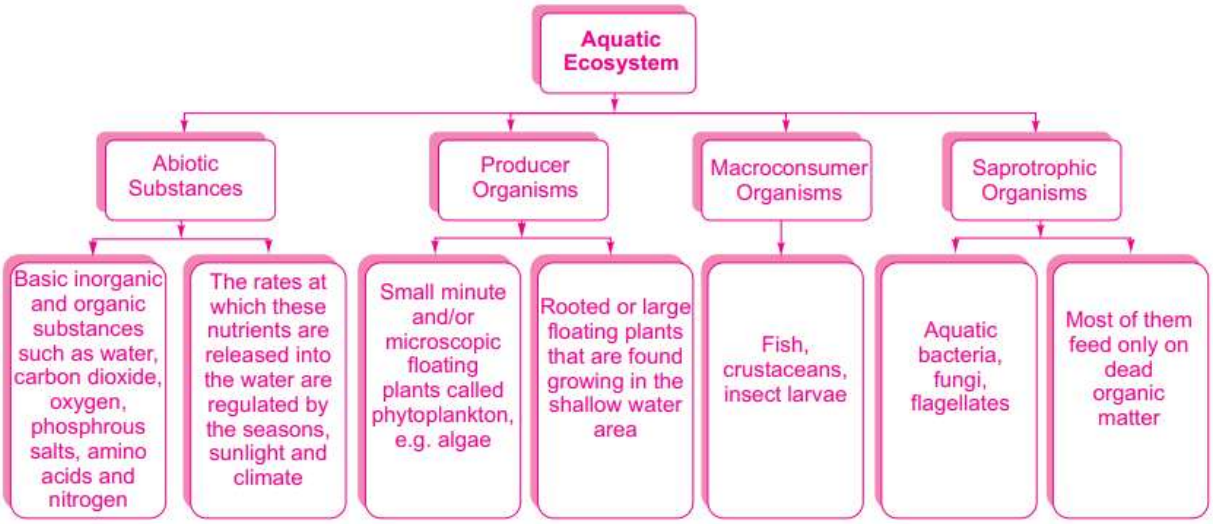
- Insects, e.g. locust, ants, spiders and scorpions
- Reptiles, e.g. snakes and lizards
- Birds, e.g. sand grouse, roadrunner, ostriches
- Mammals, e.g. hamsters, rats, kangaroo, wild horses, foxes and jackals, lions, camels, oryx.

Decomposers

Due to poor vegetation, the amount of dead organic matter is very less. As a result, the decomposers are very few. The common decomposers are some bacteria and fungi, most of which are thermophile.

Aquatic Ecosystems

Aquatic ecosystems exist in ponds, lakes, rivers, wetlands, etc. The components of an aquatic ecosystem are very diverse, but can be divided into several basic units.



Wetland Ecosystems

Structure:
Biotic Components: Wetlands are rich in biodiversity, including aquatic plants, fish, amphibians, birds, and invertebrates. Vegetation typically includes reeds, grasses, and water-tolerant trees like mangroves.

Abiotic Components: Wetlands are characterized by saturated soils, standing or slow-moving water, and a high level of nutrients. They can be freshwater or saltwater and are often found in floodplains or coastal areas.

Function:
Wetlands act as natural water filters, trapping pollutants and sediments.
They provide critical habitat for many species, particularly migratory birds.
Wetlands help in flood control, groundwater recharge, and maintaining water quality.

River Ecosystems
Structure:
Biotic Components: Rivers are home to diverse aquatic life, including fish, amphibians, aquatic plants, and invertebrates. Riparian vegetation along riverbanks supports a variety of terrestrial species.

Abiotic Components: River ecosystems are defined by flowing freshwater, with varying levels of oxygen, nutrients, and sediments depending on the river's flow rate and source.

Function:

Rivers are vital for nutrient cycling, transporting nutrients from land to sea.
They support agriculture, provide drinking water, and are crucial for transportation.
Rivers also shape landscapes through erosion and sediment deposition.

Oceanic Ecosystems

Structure:
Biotic Components: Oceans host a vast array of life, from microscopic plankton to large marine mammals like whales. Coral reefs, mangroves, and kelp forests are important oceanic habitats.

Abiotic Components: Oceans cover about 70% of the Earth's surface and are characterized by saltwater, varying depths, currents, and temperature gradients. The ocean floor and the water column create distinct habitats.
Function:
Oceans play a major role in regulating the Earth's climate by absorbing and storing solar energy and carbon dioxide.
They are essential for global nutrient cycling and provide a major source of food and resources for humans.
Oceans also support global trade and influence weather patterns through ocean currents.

Lake Ecosystems
Structure:
Biotic Components: Lakes support a variety of aquatic plants, fish, amphibians, and invertebrates. The species composition varies based on the lake's nutrient levels, depth, and temperature.

Abiotic Components: Lakes are inland bodies of standing water with variable nutrient levels (eutrophic, mesotrophic, or oligotrophic), temperature stratification, and oxygen concentration.
Function:
Lakes provide freshwater for drinking, irrigation, and industrial use.
They support biodiversity and recreational activities like fishing and boating.
Lakes play a role in nutrient cycling and can act as sediment traps.

Forest resources and associated problems

1. Use and over-exploitation.
2. Deforestation.
3. Timber extraction.
4. Mining and its effects on forest.
5. Dams and their effects on forests and tribal people.

USES OF FOREST

1. Watershed protection:
- Reduce the rate of surface run-off of water.
- Prevent flash floods and soil erosion.
- Produces prolonged gradual run-off and thus prevent effects of drought.

2. Atmospheric regulation:
- Absorption of solar heat during evapo-transpiration.
- Maintaining carbon dioxide levels for plant growth.
- Maintaining the local climatic conditions.

3. Erosion control:
- Holding soil (by preventing rain from directly washing soil away).

4. Land bank:
- Maintenance of soil nutrients and structure.

5. Local use - Consumption of forest produce by local people who collect it for subsistence (Consumptive use)
- Food - gathering plants, fishing, hunting from the forest. (In the past when wildlife was Plentiful, people could hunt and kill animals for food. Now those populations of most Wildlife species have diminished; continued hunting would lead to extinction.)
- Fodder - for cattle.
- Fuel wood and charcoal for cooking, heating.
- Poles - building homes especially in rural and wilderness areas.
- Timber – household articles and construction.
- Fiber - weaving of baskets, ropes, nets, string, etc.
- Sericulture – for silk. • Apiculture - bees for honey, forest bees also pollinate crops.
- Medicinal plants - traditionally used medicines, investigating them as potential Source for new modern drugs.

6. Market use - (Productive use)
- Most of the above products used for consumptive purposes are also sold as source of income for supporting the livelihoods of forest dwelling people.

- Minor forest produce - (non-wood products): Fuel wood, fruit, gum, fiber, etc.which are Collected and sold in local markets as a source of income for forest dwellers.
- Major timber extraction - construction, industrial uses, paper pulp, etc. Timber extraction is done in India by the Forest Department, but illegal logging continues in many of the forests of India and the world.

Major causes of Deforestation:

a. Shifting cultivation : There are an estimated 300 million people living as shifting cultivators who practice slash and burn agriculture and are supported so clear more than 5 lakh ha of forests for shifting cultivation annually. In India, we have this practice of North-East and to some extent in Andhra Pradesh, Bihar and M.P. which contribute to nearly half of the forest clearing annually.

b. Fuel requirements: Increasing demands for fuel wood by the growing population in India alone has shooted up to 300-500 million tons in 2001 as compared to just 65 million tons during independence, thereby increasing the pressure on forests.

c. Raw materials for industrial use: Wood for making boxes, furniture, railway-sleepers, plywood, match boxes, pulp for paper industry etc. have exerted tremendous pressure on forests.

Plywood is in great demand for packing tea for Tea industry of Assam while fir tree wood is exploited greatly for packing apples in J & K.

d. Development projects: Massive destruction of forests occurs for various development projects like hydroelectric projects, big dams, road construction, mining etc.

e. Growing food needs: In developing countries this is the main reason for deforestation. To meet the demands of rapidly growing population, agricultural lands and settlements are created permanently by clearing forests.

f. Overgrazing: The poor in the tropics mainly rely on wood as a source of fuel leading to loss of tree cover and the cleared lands are turned into the grazing lands. Overgrazing by the cattle leads to further degradation of these lands.

g. Conversion of forests and woodlands to agricultural land to feed growing numbers of people

Major activities and threats to Forests resources:

1.Timber Extraction: Logging for valuable timber, such as teak and Mahogany not only involves a few large trees per hectare but about a dozen more trees since they are strongly interlocked with each other a by vines etc. Also road construction for making approach to the trees causes further damage to the forests.

The steps in timber extraction are: a) Clear felling b) Mechanized logging c) Manual logging d) Selective logging

2. Mining: Mining operations for extracting minerals and fossil fuels like coal often involves vast forest areas. Mining from shallow deposits is done by surface mining while that from deep deposits is done by sub-surface mining

Sustainability and the 17 Sustainable Development Goals (SDGs)

The Sustainable Development Goals (SDGs) are a collection of 17 global goals set by the United Nations in 2015 as part of the 2030 Agenda for Sustainable Development. They are a universal call to action to end poverty, protect the planet, and ensure that all people enjoy peace and prosperity by 2030.

History of the SDGs
The SDGs were born out of the success and limitations of the Millennium Development Goals (MDGs), which were established in 2000 and targeted various issues like poverty, hunger, and disease. While the MDGs made significant progress, they were criticized for being too narrow and for not addressing the root causes of poverty or the environmental challenges that threaten sustainable development.

In 2012, during the United Nations Conference on Sustainable Development (Rio+20), member states agreed to develop a new set of goals that would be more comprehensive and universally applicable. After extensive consultations involving governments, civil society, and other stakeholders, the SDGs were adopted by all UN member states in September 2015.

The 17 SDGs and Their Targets
Each of the 17 SDGs has specific targets and indicators to measure progress. In total, there are 169 targets associated with the goals. Here is a brief overview of the SDGs:

1. No Poverty: End poverty in all its forms everywhere.
Targets: Eradicate extreme poverty, reduce poverty by at least 50%, implement social protection systems, ensure equal rights to economic resources, etc.

2. Zero Hunger: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture.
Targets: End hunger and malnutrition, double agricultural productivity, ensure sustainable food production systems, maintain genetic diversity in food production, etc.

3. Good Health and Well-being**:** Ensure healthy lives and promote well-being for all at all ages.
Targets: Reduce maternal mortality, end preventable deaths of newborns and children under 5, combat epidemics, reduce non-communicable diseases, promote mental health, etc.

4. Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
Targets: Ensure all children complete free, equitable, and quality primary and secondary education, increase the number of youth and adults with relevant skills, eliminate gender disparities in education, etc.

5. Gender Equality**:** Achieve gender equality and empower all women and girls.
Targets: End discrimination and violence against women and girls, ensure equal participation and opportunities, eliminate harmful practices, recognize and value unpaid care work, etc.

6. Clean Water and Sanitation: Ensure availability and sustainable management of water and sanitation for all.
Targets: Achieve universal and equitable access to safe and affordable drinking water, improve water quality, increase water-use efficiency, implement integrated water resources management, etc.

7. Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable, and modern energy for all.
Targets: Ensure universal access to modern energy services, increase the share of renewable energy, double the rate of improvement in energy efficiency, etc.

8. Decent Work and Economic Growth: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.
Targets: Sustain per capita economic growth, achieve higher levels of economic productivity, promote policies that support productive activities, improve global resource efficiency, etc.

9. Industry, Innovation, and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.
Targets: Develop quality, reliable, and sustainable infrastructure, promote inclusive and sustainable industrialization, enhance scientific research, and upgrade technological capabilities, etc.

10. Reduced Inequality: Reduce inequality within and among countries.
Targets: Achieve and sustain income growth of the bottom 40% of the population, empower and promote social, economic, and political inclusion, ensure equal opportunity and reduce inequalities of outcome, etc.

11. Sustainable Cities and Communities: Make cities and human settlements inclusive, safe, resilient, and sustainable.
Targets: Ensure access to adequate, safe, and affordable housing, provide access to safe, affordable, and sustainable transport systems, reduce the adverse environmental impact of cities, etc.

12. Responsible Consumption and Production: Ensure sustainable consumption and production patterns.
Targets: Implement the 10-Year Framework of Programmes on Sustainable Consumption and Production, achieve the sustainable management and efficient use of natural resources, reduce waste generation, encourage companies to adopt sustainable practices, etc.

13. Climate Action: Take urgent action to combat climate change and its impacts.
Targets: Strengthen resilience and adaptive capacity to climate-related hazards, integrate climate change measures into national policies, improve education and awareness on climate change, etc.

14. Life below Water: Conserve and sustainably use the oceans, seas, and marine resources for sustainable development.
Targets: Reduce marine pollution, protect and restore ecosystems, minimize ocean acidification, regulate harvesting and end overfishing, conserve coastal and marine areas, etc.

15. Life on Land: Protect, restore, and promote sustainable use of terrestrial ecosystems, manage forests sustainably, combat desertification, halt and reverse land degradation, and halt biodiversity loss.
Targets: Ensure the conservation, restoration, and sustainable use of terrestrial and inland freshwater ecosystems, combat desertification, halt deforestation, end poaching and trafficking of protected species, etc.

16. Peace, Justice, and Strong Institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels.
Targets: Reduce violence and related deaths, end abuse, exploitation, trafficking, and all forms of violence against children, promote the rule of law and ensure equal access to justice, reduce corruption, etc.

17. Partnerships for the Goals: Strengthen the means of implementation and revitalize the global partnership for sustainable development.
Targets: Mobilize financial resources for developing countries, enhance international support for implementing effective and targeted capacity-building, promote a universal, rules-based, open, non-discriminatory, and equitable trading system, etc.

Implementation of the SDGs

The implementation of the SDGs involves a multi-stakeholder approach, including governments, civil society, the private sector, and international organizations. Each country is responsible for incorporating the SDGs into its national policies, plans, and programs.

Key Steps in Implementation:

1. National and Local Integration: Countries need to integrate the SDGs into their national development strategies and policies, ensuring that all relevant sectors and stakeholders are engaged.

2. Monitoring and Reporting: Countries must develop indicators and systems to monitor progress toward each goal. The United Nations provides a global framework for reporting, but countries are encouraged to develop context-specific indicators.

3. Financing: Implementing the SDGs requires significant financial resources. This involves mobilizing domestic resources, attracting private investment, and accessing international funding mechanisms.

4. Innovation and Technology: Leveraging innovation and technology is essential for achieving the SDGs, particularly in areas like renewable energy, sustainable agriculture, and healthcare.

5. Public Awareness and Education: Raising awareness and educating the public about the SDGs is critical for ensuring broad-based support and participation.

Capacity Development

Capacity development is crucial for the successful implementation of the SDGs. It involves strengthening the skills, knowledge, and systems of individuals, organizations, and societies to effectively implement and sustain the SDGs.

Key Aspects of Capacity Development:

1. Institutional Capacity: Building strong institutions that can coordinate, monitor, and implement SDG-related initiatives.

2. Human Capacity: Training and educating individuals in both the public and private sectors on sustainable development practices.

3. Technical Capacity: Enhancing the ability to collect, analyze, and use data for decision-making, as well as promoting the use of technology to achieve the SDGs.

4. Financial Capacity: Developing mechanisms to mobilize and manage the financial resources needed to achieve the SDGs.

5. Community Engagement: Strengthening the capacity of local communities to participate in and benefit from SDG-related activities.

The 17 SDGs represent a comprehensive framework for addressing the most pressing global challenges. Their successful implementation depends on coordinated efforts at all levels of society, from local communities to global partnerships.

Multiple Choice Questions (MCQ)

1. Which of the following statement is true
   a. Green plants are self-nourishing b. Producers depends on consumers
   b. Biotic components includes all non-living d. Herbivores depend on Carnivores

2. Primary consumer is
   a. Herbivores b. Carnivores c. Macro consumers d. Omnivores

3. A predator is
   a. An animal that is fed upon b. **An animal that feeds upon another animal**
   c. Animal that feeds upon both plants and animals d. A primary consumer

4. Access to food is mainly determined by
a) Household income b) Food assistance programmes
c) Human resources d) Society/Community

5. The word 'Environment' is derived from
a) Greek b)French c) Spanish d) English

6. Which among the following is a climatic factor?
a) pressure b) humidity c) temperature **d) all of the above**

7. The major atmospheric gas layer in stratosphere is
a) Hydrogen b) Carbon dioxide **c) Ozone** d)Helium

8. Which atmospheric sphere is closest to the earth surface?
a) Troposphere b) stratosphere c) mesosphere d) Exosphere

9. Which of the following is the terrestrial ecosystem?
a) forest b) grass land c) desert **d) all of the above**

10. Ecological pyramids are studies of
a) Pyramid of numbers b) pyramid of biomass c) pyramid of energy **d) all of the above**

11. World Environment day is on
a) 5th May **b) 5th June** c) 18th July d) 16th August

12. Factors responsible for balanced ecosystem are
a) balance between predator and prey
b) balance between vegetation, herbivorous and carnivorous
c) balance between competing species and biotic factors
d) all of the above

13. Which of the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide
b) Water
c) Nutrients
d) All of above

14. Habitat refers to
    a) Physical conditions of the place where organisms live
    b) Chemical conditions of the place where organisms live
    c) Both a & b
    d) None of a or b

15. Essential component of social security are
    a) Meeting personal growth and development
    b) Maintaining natural capital
    c) Fairness and equity in distribution of costs of resources.
    d) Community resilience

16. Socio – economic security in environmental aspects involves
    a) Fairness & equity in distribution costs for complete existing generation
    b) Welfare of the present generation
    c) Intra and inter generational equity of resources
    d) All of the above

17. A food web consists of
    a) a portion of a food chain c) an organisms position in a food chain
    b) interlocking food chains d) a set of similar consumers

18. Which of the following statements are true?
    a) Man is not dependent on nature
    b) Resources are unlimited, so one can use them as per one's wish.
    c) Energy can be converted from one form to another, but some percentage is lost into the environment
    d) Matter can be generated afresh. It need not be recycled or reused.

19. Which of the following conditions must be fulfilled to ensure food security?
    a) Food must be available
    b) Each person must have access to it
    c) Food utilized/consumed must fulfill nutritional requirements
    d) all of the above

20. 'Earth Day' is held every year on
    a) June 5th b) November 23rd **c) April 22nd** d) January 26th

21. Eutrophication is
    a) an improved quality of water in lakes
    b) a process in carbon cycle
    c) the result to accumulation of plant nutrients in water bodies
    d) a water purification technique

22. Major purpose of most of the Dams around the world is
    a) Power generation b) Drinking water supply c) Flood control **d) Irrigation**

23. The term 'Environment' has been derived from the French word ….which means to encircle or surround
    a. **Environ** b) Oikos c) Geo d) Aqua

24. The objective of environmental education is
    a) Raise consciousness about environmental conditions
    b) To teach environmentally appropriate behaviour
    c) Create an environmental ethic
    d) All of the above

25. Which of the following conceptual spheres of the environment is having the least storage capacity for matter?
a) Atmosphere b) Lithosphere c) Hydrosphere d) Biosphere

26. Which of following components of environment are effective transporters of matter?
a) Atmosphere and Hydrosphere b) Atmosphere and Lithosphere
c) Hydrosphere and Lithosphere d) Biosphere and Lithosphere

27. Biosphere is
    a. The solid shell of inorganic materials on the surface of the earth
    b. The thin shell of organic matter on the surface of earth comprising of all the living things
    c. the sphere which occupies the maximum volume of all of the spheres
    d. all of the above

28. Atmosphere consists of 79 per cent Nitrogen and 21 per cent Oxygen by
a. volume b) weight c) Density d) All the three

29. Which of the following is a biotic component of an ecosystem?
a. Fungi b) solar light c) temperature d) humidity

30. In an ecosystem, the flow of energy is
a) Bidirectional b) Cyclic **c) Unidirectional** d) Multidirectional

31. Which Pyramid is always upright
a. energy b) biomass c) numbers d) food chain

32. In complex ecosystems the degree of species diversity is
a. poor **b) high** c) medium d)none

33. The organisms who directly feed on producers are called
a. Herbivores b) Carnivores c) Decomposers d) Saprophytes

34. The sequence of eating and being eaten in an ecosystem is called
a. Food Chain b) carbon cycle c) hydrological cycle d) anthroposystem

35. Which of the following is a producer in an ecosystem
    a) Plants and some bacteria capable of producing their own food
    b) Animals
    c) Human beings
    d) Fish

36. Which of the following statements is false?
    a) Inorganic nutrients are recycled in an ecosystem
    b) Energy 'flows' through the ecosystem in the form of carbon-carbon bonds
    c) Energy is recycled in an ecosystem
    d) Respiration process releases energy

37. The largest reservoir of nitrogen in our planet is
    a. Oceans **b) Atmosphere** c) biosphere d) Fossil fuels

38. In aquatic ecosystem phytoplankton can be considered as a
    a. Consumer **b) producer** c) Saprotrophic organisms d) Macroconsumer

39. The basic requirements of human beings are provided by
    a. Industrialisation
    b. Agriculture
    c. Nature
    d. Urbanisation

40. Environment is the life support system that includes
    a. Air
    b. Water
    c. Land
    d. All of the above

41. In an ecosystem biological cycling of materials is maintained by
    a. Producer
    b. Consumer
    c. Decomposer
    d. All of the above

42. Organisms which feed directly or indirectly on producers are called
    a. Prey
    b. Consumers
    c. Decomposers
    d. Detritus

43. The primary producers in a forest ecosystem are
    a. Chlorophyll containing trees and plants
    b. Herbivores
    c. Carnivores
    d. Bacteria and other microorganisms

44. Abiotic component includes
    a. Soil
    b. Temperature
    c. Water
    d. All of the above